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3.1) Determine the z transform, including the ROC.

3.1-c) $\left(\frac{1}{2}\right)^n u[-n]$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^n u[-n]\, z^{-n}$$

$$= \sum_{n=-\infty}^{0} \left(\frac{1}{2}\right)^n z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^{-n} z^{n} = \sum_{n=0}^{\infty} (2z)^n = \frac{1}{1-2z} \qquad |z| < \frac{1}{2} \quad : \text{ROC}$$

3.1-g) $\left(\frac{1}{2}\right)^n (u[n] - u[n-10]) = x[n]$

★ $\sum_{k=N_1}^{N_2} a^k = \frac{a^{N_1} - a^{N_2+1}}{1-a}$

$$x[n] = \left(\frac{1}{2}\right)^n (u[n] - u[n-10])$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n} = \sum_{n=0}^{9} \left(\frac{1}{2}\right)^n z^{-n} = \sum_{n=0}^{9} \left(\frac{1}{2z}\right)^n = 1 + \frac{1}{2z} + \left(\frac{1}{2z}\right)^2 + \cdots + \left(\frac{1}{2z}\right)^9$$

$$= \frac{1-\left(\frac{1}{2z}\right)^{10}}{1-\frac{1}{2z}} = \frac{1-(2z)^{-10}}{1-(2z)^{-1}} \qquad \text{ROC: } |z| > 0$$

3.4)

a) ROC: $\frac{1}{3} < |z| < 2$ because the Fourier transform exists the unit circle have to be included to the ROC. The ROC can't have (include) poles.

According to the ROC $x[n]$ is two sided

b) There is 2 possible sequence $\frac{1}{3} < |z| < 2$ and $2 < |z| < 3$ because ROC can't have (include) poles.

c) To be casual the ROC have to be right sided and to be stable it have to include the unit circle but it is not possible at the same time for the given pole-zero plot.

3.4. Consider the z-transform $X(z)$ whose pole–zero plot is as shown in Figure P3.4.

(a) Determine the ROC of $X(z)$ if it is known that the Fourier transform exists. For this case, determine whether the corresponding sequence $x[n]$ is right sided, left sided, or two sided.

(b) How many possible two-sided sequences have the pole–zero plot shown in Figure P3.4?

(c) Is it possible for the pole–zero plot in Figure P3.4 to be associated with a sequence that is both stable and causal? If so, give the appropriate ROC.

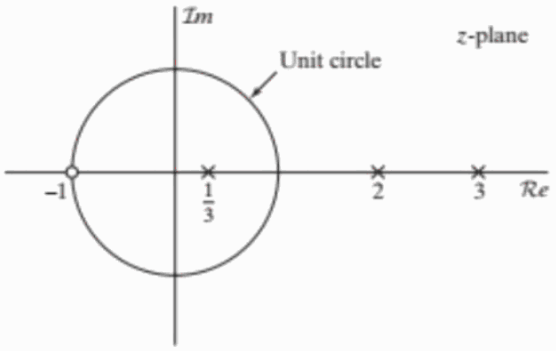


Figure P3.4

a) ROC : $\frac{1}{3} < |z| < 2$ because the Fourier transform exists the unit circle have to be included to the ROC. The ROC can't have (include) poles.

According to the ROC $x[n]$ is two sided

b) There is 2 possible sequence $\frac{1}{3} < |z| < 2$ and $2 < |z| < 3$ because ROC can't have (include) poles.

c) To be casual the ROC have to be right sided and to be stable it have to include the unit circle but it is not possible at the same time for the given pole-zero plot.